Class 12: Maxwell-Boltzmann Statistics

From the discussions in the earlier classes we have noted that the Drude model needs to be improved. In particular, there appears to be an incorrect assumption of how energy is distributed amongst the free electrons in the Drude model, which led to incorrect estimations of C_V^e and $\langle v^2 \rangle$. The kind of system we are dealing with is extremely large in the sense that it consists of 10^{28} particles per cubic meter. This is too large a number to compute averages by determining the properties of each and every particle – instead we will have to use statistical approaches to handle problems of this kind. Statistical approaches for problems of this kind recognize that such systems have a 'most probable state'. With very large system the most probable state of the system is more probable than all of the other states of the system combined. In such large systems, the most probable state overwhelms the probability of occurrence of all other states combined, i.e. the probability of occurrence of the most probable state is nearly 100%. Therefore, in large systems, the most probable state is effectively the equilibrium state, the state in which it will be found almost all of the time, and hence the only state worth further consideration. Statistical mechanics takes advantage of this situation, and while the mathematics to calculate all possible states may be quite laborious, statistical mechanics ignores all of the other states and focuses on identifying only the most probable state, which is significantly easier to do.

For the particles we are dealing with, free electrons in a solid, the Drude model has made the assumption that their behavior is similar to that of atoms of an ideal gas. Specifically, this means that the electrons behave as though they are 'identical but distinguishable'. The words 'identical but distinguishable' may convey something in a conversational sense but they mean something specific in the context of statistics of particles. In this class we will not dwell on the significance of 'identical but distinguishable' but merely note that we have made such an assumption. In the next class we will examine these words 'identical but distinguishable' and examine what they mean, and consider what other possibilities and variations exist in this context.

The statistical distribution of properties of a large collection of identical but distinguishable particles has been calculated and is attributed to Maxwell and Boltzmann, and hence bears their names. By assuming that the free electrons in a solid behave like atoms of an ideal gas, the Drude model has assumed that the free electrons follow the Maxwell-Botzmann statistics.

In the analysis that follows, we will initially talk of all possible states and initiate the mathematics towards that goal, but at an appropriate step, we will narrow down to the most probable state, and from there on focus only on the most probable state and its properties.

Consider a system that has a total energy 'U', a total volume 'V' and a total number of particles 'n', all of which are constant and together represent the macrostate of the system. Let the system have allowed energy levels $\varepsilon_0, \varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4, \ldots \varepsilon_r$

We seek the information, how many particles, or what fraction of particles, are in a particular energy level. In other words our goal is to identify the distribution of particles corresponding to the most probable microstate of the system

In a given microstate corresponding to the above macro state, let there be n_0 particles at ε_0, n_1 particles at ε_1, n_2 particles at ε_2, n_3 particles at ε_3, n_4 particles at ε_4,...and n_r particles at ε_r,

The number of ways in which this microstate can be accomplished is denoted by Ω which is given by:

$$\Omega = \frac{n!}{n_0!*n_1!*n_2!*n_3!*n_4!*...n_r!}$$

$\Omega = \frac{n!}{\prod_{i=0}^{i=r} n_i!}$; where the symbol $\prod$ denotes the product of the factorials.

We now take advantage of Sterling's approximation, which states that for large 'n',
$\ln n! = n \ln n - n$

Since the system is large, we assume each of the n_i, is large enough to justify the use of this approximation.

Therefore

$$\ln \Omega = n \ln n - n - \sum_{i=0}^{i=r} (n_i \ln n_i - n_i)$$

Our goal is to identify the specific distribution, i.e. the specific values of n_i, which will maximize Ω. Please note, in the system we are examining, the values of ε_i are fixed. Only the values of n_i, at each of the ε_i can be varied. In other words, we wish to know what is the value of n_0 at ε_0, n_1 at ε_1, n_2 at ε_2, etc. such that the combination of n_0, n_1, n_2, etc, maximizes Ω.

To figure out the combination that maximizes Ω, we make note of the fact that by the nature of the behavior of Ω and $\ln\Omega$, the combination that maximizes Ω is the same combination that also maximizes $\ln\Omega$.

If $\ln\Omega$ has been maximized, its differential with respect to n_i will equal zero. Another way of stating this is that for a small rearrangement of particles between the different states, $\delta\ln\Omega = 0$

Therefore differentiating the expression for $\ln\Omega$ with respect to n_i, we get:

$$\sum_{i=0}^{i=r} \partial n_i \ln n_i + \partial n_i - \partial n_i = 0$$

or

$$\sum_{i=0}^{i=r} \partial n_i \ln n_i = 0 \qquad (1)$$

In addition, since the total number of particles is fixed, any rearrangement of particles does not change the total number of particles.

Therefore,

$$\sum_{i=0}^{i=r} \partial n_i = 0 \qquad (2)$$

Finally, since the total energy of the system is fixed, rearrangement of particles between the different states cannot change the total energy of the system

Therefore,

$$\sum_{i=0}^{i=r} \varepsilon_i \partial n_i = 0 \qquad (3)$$

Our problem can therefore be restated as follows, we wish to maximize $\ln\Omega$, as denoted by equation (1), subject to the constraints imposed by equations (2) and (3).

Mathematically, this problem is identical to trying to find the maximum of a surface, subject to the constraint that you have to move along a specific path. In other words, we are not trying to find the absolute maximum of the surface, but merely the highest point in the surface that can reached while constrained to moving along the specific path.

The mathematical approach used to accomplish the above is referred to as the Lagrange method of undetermined multipliers. The approach and idea behind the method is as follows: firstly the three equations are rolled into one by multiplying equation (2) by a presently unknown multiplier α, and equation (3) by a multiplier β, and adding the three equations.

$$\sum_{i=0}^{i=r} \partial n_i (\ln n_i + \alpha + \beta \varepsilon_i) = 0 \qquad (4)$$

The method assumes that we are able to find α and β such that the various δn_i, become independent of each other. In other words, upon reaching a maximum, the result should not change due to minor variations of specific n_i and therefore should be independent of the various δn_i

The implication of δn_i, becoming independent of each other is that the term within the parenthesis in equation (4) now equals zero.

Therefore:

$$\ln n_i + \alpha + \beta \varepsilon_i = 0$$

which implies that

$$n_i = e^{-\alpha} e^{-\beta \varepsilon_i}$$

In other words, our result indicates that maximizing $\ln\Omega$, occurs when there is an exponential decrease in n_i, as ε_i increases.

Since $\sum_{i=0}^{i=r} n_i = n$

$$e^{-\alpha} \sum_{i=0}^{i=r} e^{-\beta \varepsilon_i} = n$$

the term $\sum_{i=0}^{i=r} e^{-\beta \varepsilon_i}$ is called the partition function 'P'.

Therefore:

$$e^{-\alpha} = \frac{n}{P}$$

further, the value of β works out to be $\frac{1}{k_b T}$; where k_b is the boltzmann constant, and T is the absolute temperature.

Hence:

$n_i = \frac{n}{P} e^{-\frac{\varepsilon_i}{k_b T}}$; an exponential decrease in n_i, as ε_i increases. This expression is referred to as the Maxwell-Boltzmann distribution.

Figure 12.1below shows a plot of the Maxwell-Boltzmann statistics. The energy levels available in the system are plotted on the y-axis, while the number of allowed states at a given energy level, are plotted on the x-axis

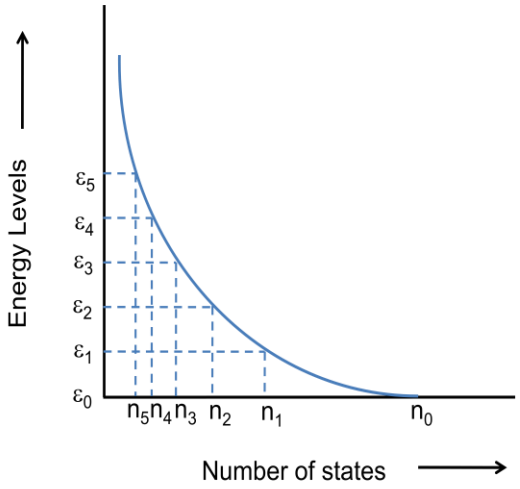


Figure 12.1: A plot showing the number of states as a function of energy level, as determined by the Maxwell-Boltzmann statistics.

As seen from the plot above, the higher the energy level, the lower the number of particles present.

Atoms of an ideal gas, and electrons in the free electron gas when described by the classical Drude model, are assumed to conform to the above distribution.